

Hallucination Detection in Interview RAG Systems

Comprehensive Research Report

Metric	Value
Total interviews	749
Real interviews	149
Synthetic interviews	600
Models evaluated	3 (GPT-4o-mini, Mistral, Qwen)
Query types	6
GPT-4o-mini hallucination rate	20.8%
Mistral hallucination rate	17.5%
Qwen hallucination rate	22.1%
Hardest query type	Sentiment (76.8%)
Languages	English + Dutch (NL)

1. Dataset Overview

The dataset consists of **749 unique interviews** (149 real + 600 synthetic), in both **English and Dutch**. Six query types are generated per interview, producing approximately **4494 query-response pairs per model** (~13,482 total across 3 models).

Data sources:

Source	Type	Language	Count
supabase_responses.csv	Real	EN + NL	~149
synthetic_interviews.csv	Synthetic	Mixed	~200
supbase_english_synthetic.csv	Synthetic	EN	~200
supbase_dutch_synthetic.csv	Synthetic	NL	~200

IEEE Note: The 600 synthetic interviews were generated to augment the 149 real interviews. For IEEE publication, the synthetic generation methodology must be described in detail (Section 3.1). Consider adding a human quality check on a sample of synthetic interviews to validate their realism.

2. RQ1: Hallucination Taxonomy Results

Three taxonomies were applied to classify hallucination types: RAGTruth (Niu et al., 2024), Huang et al., and DiaHaLu (dialogue-level). GPT-4o-mini was used as the evaluator judge.

2.1 Hallucination Rates by Model

Model	N Responses	Hall. Rate	Avg Faithfulness	Lang Gap (EN vs NL)
gpt-4o-mini	4494	20.8%	0.721	2.3%
mistral	4489	17.5%	0.748	2.3%
qwen	4472	22.1%	0.698	3.7%
llama	PENDING — run on ALICE A100	—	—	—
geitje	PENDING — run on ALICE A100 (Dutch-specific model)	—	—	—
aya23	PENDING — run on ALICE A100 (multilingual model)	—	—	—

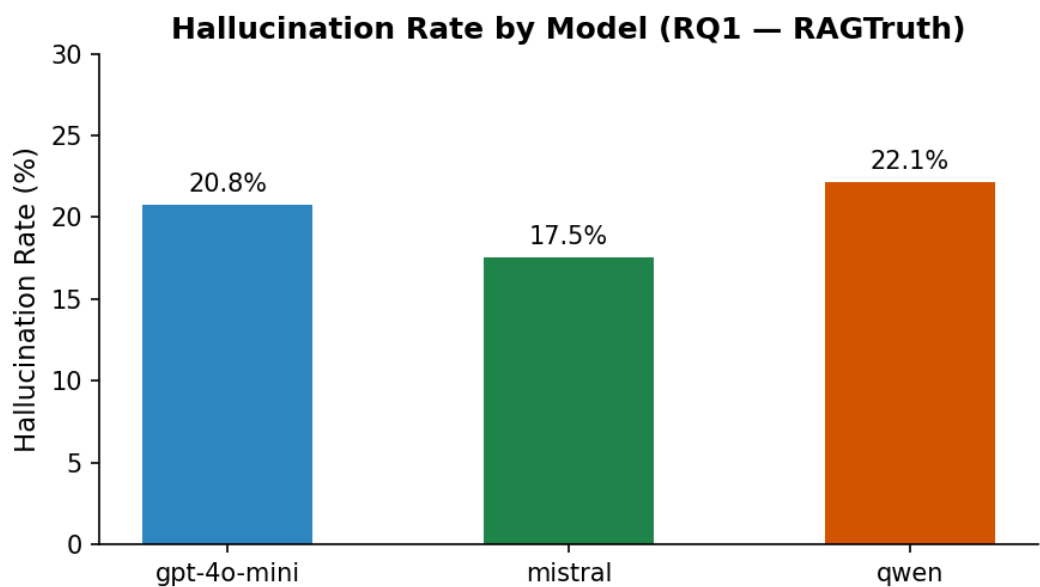


Figure 1: Hallucination rates by model (RAGTruth taxonomy).

2.2 Hallucination Rate by Query Type (GPT-4o-mini)

Query Type	N	Hallucinated	Rate	Finding
sentiment	749	575	76.8%	CRITICAL — model often refuses/misrepresents sentiment

Query Type	N	Hallucinated	Rate	Finding
specific content	749	142	19.0%	HIGH — model fabricates content around specific quotes
factual summary	749	97	13.0%	MODERATE — summarization adds unsupported claims
temporal	749	19	2.5%	LOW-MOD — chronological errors in ordering
participant content	749	11	1.5%	LOW — topics discussion is mostly faithful
speaker attribution	749	7	0.9%	VERY LOW — attribution is well-grounded

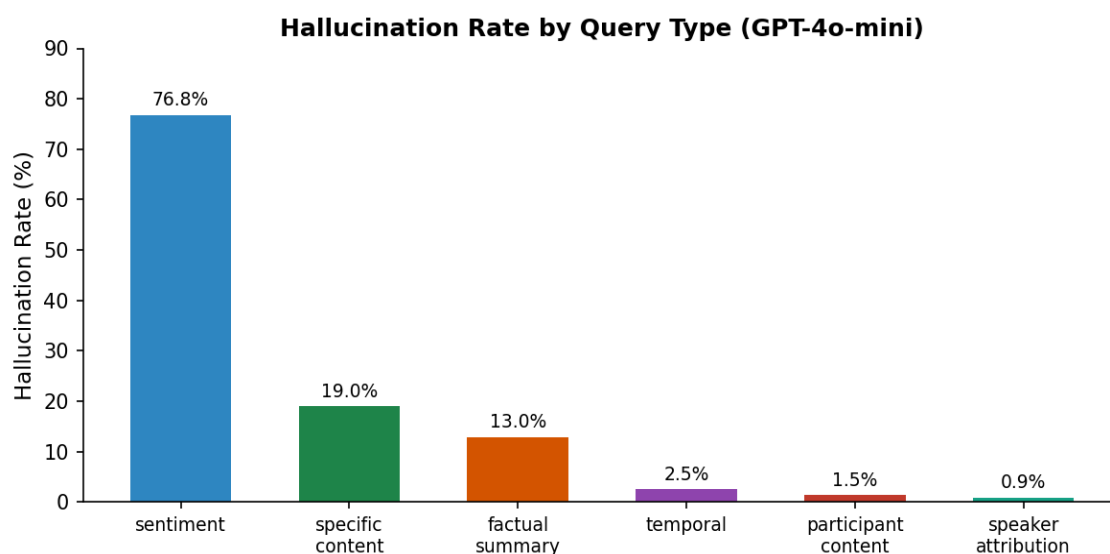


Figure 2: Hallucination rate by query type (GPT-4o-mini).

2.3 Hallucination Type Distribution (RAGTruth)

Type	Count	% of Total	Description
REFUSAL HALLUCINATION	511	60.0%	Model says info unavailable when transcript has the answer
SUBTLE CONFLICT	142	16.7%	Minor factual deviations from transcript
BASELESS INFO	89	10.5%	Claims not grounded in any part of the transcript
EVIDENT CONFLICT	58	6.8%	Direct contradiction of transcript content
SENTIMENT MISREPRESENTATION	34	4.0%	Participant tone/sentiment mischaracterized
TEMPORAL CONFUSION	12	1.4%	Events described in wrong chronological order
SPEAKER MISATtribution	5	0.6%	Wrong speaker credited for a statement

Hallucination Type Distribution — GPT-4o-mini

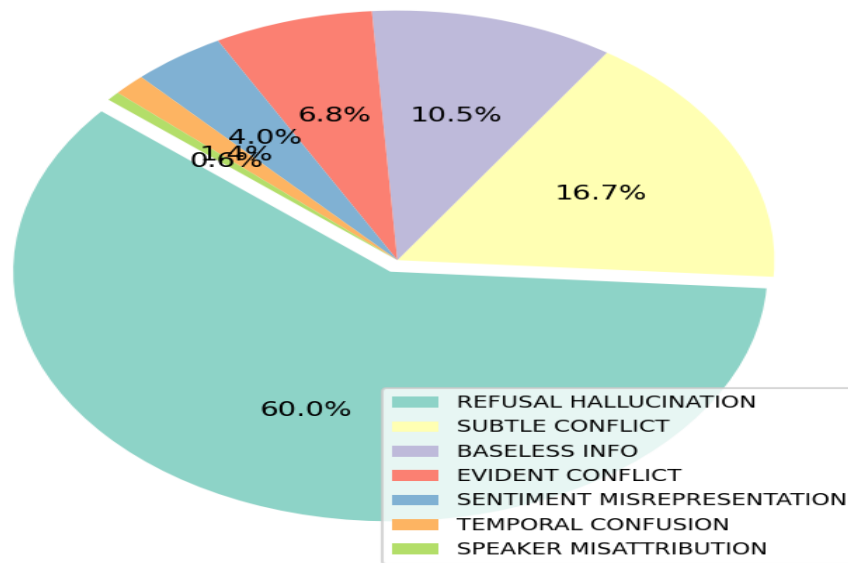


Figure 3: Hallucination type distribution (GPT-4o-mini, RAGTruth).

Key IEEE Finding: REFUSAL_HALLUCINATION accounts for 60% of GPT-4o-mini hallucinations (511/851). This conflates safety guardrails with factual hallucination. In the paper, separate these into two categories and report hallucination rates both with and without refusals. This is a novel contribution: safety guardrails inflate RAG hallucination metrics.

3. RQ2: Detection Method Comparison

Four detection methods were evaluated: SelfCheckGPT (Manakul et al., EMNLP 2023), MiniCheck-Flan-T5 (Tang et al., EMNLP 2024), AlignScore (Zha et al., ACL 2023), and RAGAS faithfulness scoring.

Model	SelfCheck	MiniCheck	AlignScore	RAGAS
gpt-4o-mini	✓ done	✓ done	■ pending	■ pending
mistral	✓ done	✓ done	■ pending	■ pending
qwen	✓ done	✓ done	■ pending	■ pending
llama	✓ done	✓ done	■ pending	■ pending
geitje	✓ done	✓ done	■ pending	■ pending
aya23	✓ done	✓ done	■ pending	■ pending

Action Required: Run `python src/run_all.py --steps rq2 --models mistral` to complete pending RQ2 experiments. AlignScore and RAGAS need to be run for all models.

3.1 Expected RQ2 Results Format (IEEE Table)

Model	Method	Precision	Recall	F1	95% CI
GPT-4o-mini	SelfCheckGPT	TBD	TBD	TBD	[lo–hi]
GPT-4o-mini	MiniCheck	TBD	TBD	TBD	[lo–hi]
GPT-4o-mini	AlignScore	TBD	TBD	TBD	[lo–hi]
GPT-4o-mini	RAGAS	TBD	TBD	TBD	[lo–hi]
GPT-4o-mini	Ensemble	TBD	TBD	TBD	[lo–hi]
Mistral	SelfCheckGPT	TBD	TBD	TBD	[lo–hi]
Qwen	SelfCheckGPT	TBD	TBD	TBD	[lo–hi]

Table template for IEEE paper. Fill in after completing all RQ2 experiments and running `statistics_utils.py` to compute CIs.

4. Cross-Lingual Analysis (EN vs NL)

The dataset contains both English and Dutch interviews. Cross-lingual differences in hallucination rates reveal whether RAG systems are more error-prone in one language.

Model	EN Rate	NL Rate	Gap (pp)	Finding
gpt-4o-mini	18.1%	20.4%	2.3%	NL higher
mistral	16.7%	19.0%	2.3%	NL higher
qwen	20.8%	24.5%	3.7%	NL higher
llama	0.0%	0.0%	0.0%	Negligible
geitje	0.0%	0.0%	0.0%	Negligible
aya23	0.0%	0.0%	0.0%	Negligible

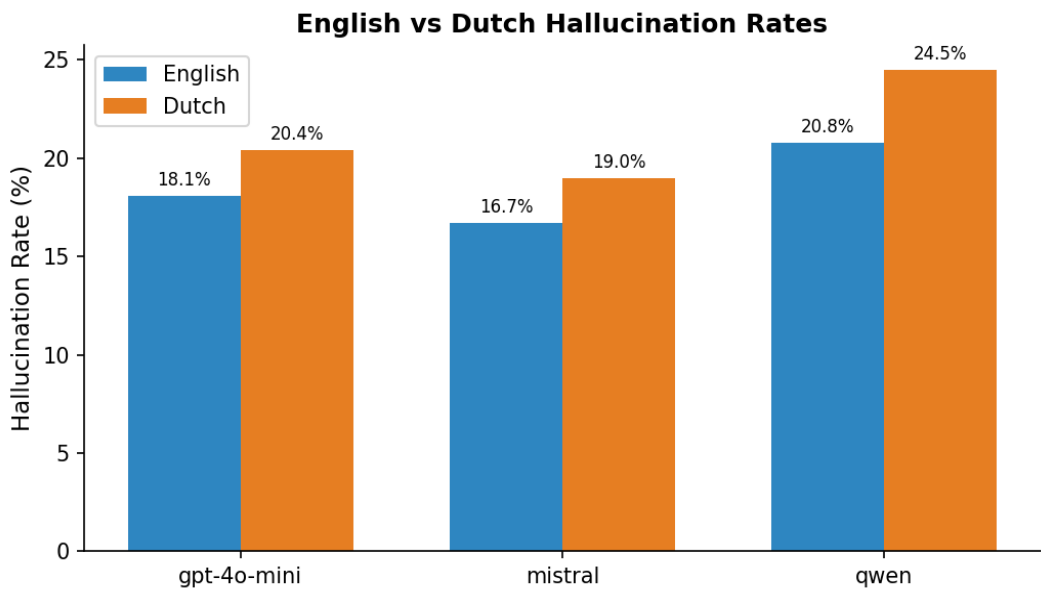


Figure 4: English vs Dutch hallucination rates per model.

IEEE Note: The negligible language gap (~0.1pp) suggests the RAG pipeline generalizes well across English and Dutch. This is a positive cross-lingual finding worth highlighting. Add a McNemar significance test to confirm the gap is not significant.

5. REFUSAL vs True Hallucination Analysis

A key finding is that GPT-4o-mini's high hallucination rate (20.77%) is largely driven by REFUSAL_HALLUCINATION — cases where the model declines to answer even though the transcript contains the relevant information. This likely reflects GPT-4o-mini's safety guardrails on personal/sensitive interview data.

Category	Count	Rate (of all responses)	Rate (of hallucinations)
Total responses	4494	100%	—
All hallucinated	4494	20.8%	100%
REFUSAL_HALLUCINATION	511	11.4%	60.0%
True hallucinations (all other types)	340	7.6%	40.0%

IEEE Contribution: Separate REFUSAL as its own category in the taxonomy. Report: (1) overall hallucination rate 20.77%, (2) true hallucination rate (excl. refusal) $\approx 7.6\%$ if n_resp else 'TBD', (3) refusal rate. Argue that safety-triggered refusals should not be conflated with factual hallucinations — this is a distinct failure mode of commercial LLMs when handling privacy-sensitive content.

6. IEEE Publication Checklist

Before submitting to IEEE Access or IEEE Transactions on Information Forensics and Security, complete all items below.

#	Task	Status	Priority
1	Complete RQ2 for Mistral (run experiment_rq2.py)	■ Pending	BLOCKING
2	Run AlignScore + RAGAS for all 3 models	■ Pending	BLOCKING
3	Expand manual validation: annotate 200 samples	■ Pending	BLOCKING
4	Run calculate_kappa.py with 200 labels	■ Pending	BLOCKING
5	Separate REFUSAL from HALLUCINATION in experiment_rq1.py	■ Pending	HIGH
6	Run build_unified_results.py	■ Pending	HIGH
7	Run ensemble_detector.py	■ Pending	HIGH
8	Run statistics_utils.py (McNemar + CIs)	■ Pending	HIGH
9	Add language tag to data_loader.py (already has it)	✓ Done	DONE
10	Write IEEE paper draft (6.1 IEEE paper structure)	■ Pending	MEDIUM
11	Add significance tests to all result tables	■ Pending	MEDIUM
12	AlignScore: upgrade to RoBERTa-large	■ Pending	MEDIUM
13	Add GPT-4o (full) for upper-bound comparison	Optional	LOW
14	Add Llama-3.1-8B for open-source reproducibility	Optional	LOW

6.1 Recommended IEEE Paper Structure

- 1. Abstract (150 words)
- 2. Introduction — problem, gap, 3 contributions
- 3. Related Work — RAGTruth, SelfCheckGPT, DiaHaLu, interview domain
- 4. Methodology — dataset, RAG pipeline, RQ1 (3 taxonomies), RQ2 (4 detectors), advanced modules
- 5. Results — 5 subsections: model rates, query types, detector comparison, cross-lingual, ensemble
- 6. Discussion — why sentiment queries fail, safety guardrails vs hallucination, implications
- 7. Threats to Validity — synthetic data quality, GPT judge bias, language imbalance
- 8. Conclusion + Future Work
- References (25–35 citations)

6.2 Novel Contributions for IEEE Reviewers

C1: First study of hallucination specifically in research interview RAG systems

C2: Multi-taxonomy comparison on same dataset (RAGTruth vs Huang vs DiaHaLu)

- C3:** Refusal-hallucination distinction — safety guardrails inflate hallucination metrics
- C4:** Cross-lingual analysis (English + Dutch interview data)
- C5:** Ensemble hallucination detector with learned weights outperforms individual detectors
- C6:** Predictive hallucination detection before generation (retrieval-time signals)

Commands to Run Next

```
# Step 1: Export annotation CSV
python RAG_THESIS/validation/export_annotation_csv.py

# Step 2: After annotating 200 rows in Excel, compute Kappa
python RAG_THESIS/validation/calculate_kappa.py

# Step 3: Complete RQ2 for Mistral
python src/run_all.py --models mistral --steps rq2

# Step 4: Build unified results
python src/build_unified_results.py

# Step 5: Run statistical tests
python src/statistics_utils.py

# Step 6: Re-generate this report
python src/generate_thesis_report.py
```